

Name: _____

Date: _____

VOLUME AND SURFACE AREA, (VOLUME OF PYRAMIDS, CONES AND SPHERES)

LO: I can calculate the volume of pyramids, cones and spheres.

Volume formulae

Pyramid $V = \frac{1}{3} \times \text{base area} \times \text{height}$. **Cone** $V = \frac{1}{3}\pi r^2 h$.

Sphere $V = \frac{4}{3}\pi r^3$.

Key pattern: Pyramids and cones are $\frac{1}{3}$ of the equivalent prism/cylinder.

Key formulae

●	Cone $V = \frac{1}{3}\pi r^2 h$	third of cylinder
●	Sphere $V = \frac{4}{3}\pi r^3$	four-thirds pi r cubed
●	Pyramid $V = \frac{1}{3} \times \text{base} \times h$	third of prism
●	Cone $V = \pi r^2 h$	that is cylinder volume
●	Sphere $V = 4\pi r^3$	must be $\frac{4}{3}\pi r^3$

Key idea

Pyramids and cones = $\frac{1}{3}$ of the prism/cylinder. Sphere = $\frac{4}{3}\pi r^3$.

Cone: $\frac{1}{3}\pi r^2 h$ | **Sphere:** $\frac{4}{3}\pi r^3$

Remember the $\frac{1}{3}$ for pointed shapes

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - cone

Find the volume of a cone with $r=3$, $h=8$.

$$\begin{aligned} V &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3} \times \pi \times 9 \times 8 \\ &= 24\pi = 75.4 \text{ cm}^3 \end{aligned}$$

75.4 cm³

One-third of the equivalent cylinder.

Example 2 - sphere

Find the volume of a sphere with $r=6$.

$$\begin{aligned} V &= \frac{4}{3}\pi r^3 \\ &= \frac{4}{3} \times \pi \times 216 \\ &= 288\pi = 904.8 \text{ cm}^3 \end{aligned}$$

904.8 cm³

Cube the radius, multiply by $\frac{4}{3}$ and π .

Example 3 - pyramid

Square-based pyramid: base 10×10 , height 12.

$$\begin{aligned} \text{Base area} &= 100 \\ V &= \frac{1}{3} \times 100 \times 12 \\ &= 400 \text{ cm}^3 \end{aligned}$$

400 cm³

One-third of base area times height.

Example 4 - hemisphere

Find the volume of a hemisphere with $r=5$.

$$\begin{aligned} V &= \frac{1}{2} \times \frac{4}{3}\pi r^3 \\ &= \frac{2}{3}\pi \times 125 \\ &= \frac{250}{3}\pi = 261.8 \text{ cm}^3 \end{aligned}$$

261.8 cm³

Half of a full sphere volume.

YOUR TURN – PRACTICE (deliberate practice)

1. Cones

Find the volume (1 dp).

- a) $r=4$, $h=9$
- b) $r=5$, $h=12$
- c) $r=6$, $h=10$
- d) $r=2$, $h=15$

2. Spheres

Find the volume (1 dp).

- a) $r=3$
- b) $r=5$
- c) $r=10$
- d) $d=8$ ($r=4$)

3. Pyramids

Find the volume.

- a) Square base 6×6 , $h=9$
- b) Rectangular base 8×5 , $h=12$
- c) Triangular base area 24, $h=10$
- d) Square base 4×4 , $h=6$

4. Hemispheres

Find the volume (1 dp).

- a) $r=4$
- b) $r=6$
- c) $r=9$
- d) $d=10$ ($r=5$)

5. Mixed

Solve each (1 dp).

a) Cone: $r=3$, $h=7$

b) Sphere: $V=36\pi$. Find r .

c) Pyramid: $V=200$, base=50. Find h . d) Football $r=11$ cm. Volume?

e) Ice cream cone $r=3$, $h=10$ + hemisphere. Total V ?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Cone volume = $\frac{1}{3}\pi r^2 h$ ☐

Correct answer: _____

Reason: _____

b) Sphere volume = $4\pi r^3$ ☐

Correct answer: _____

Reason: _____

c) A pyramid is $\frac{1}{3}$ of the equivalent prism ☐

Correct answer: _____

Reason: _____

d) Doubling the radius of a sphere doubles the volume ☐

Correct answer: _____

Reason: _____

e) Hemisphere volume = $\frac{2}{3}\pi r^3$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A cone and a sphere have the same radius $r = 6$ cm. The cone has height $h = 24$ cm. Show that the cone has the same volume as the sphere.

Expression: _____

Simplified: _____

b) A solid is made of a cone ($r=5$, $h=12$) on top of a hemisphere ($r=5$). Find the total volume and total surface area.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

